Perturbative treatment of non-local operators in quantum Monte Carlo

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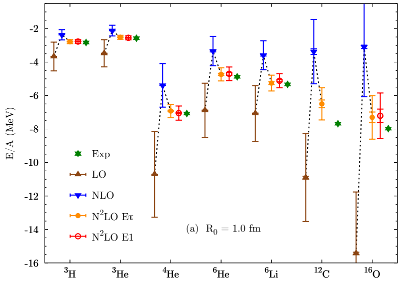
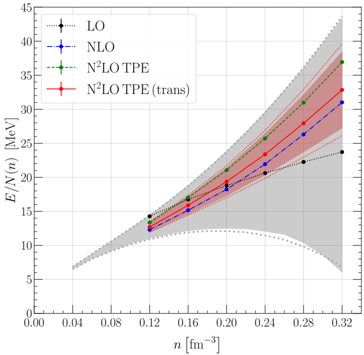
Motivation - Pushing AFDMC to N^{3}LO

- Our current QMC calculations of light nuclei and neutron matter EOS include terms only up to N^{2}LO
- There is a nonlocal "obstacle" to be dealt with first



D. Lonardoni, S. Gandolfi, J.E. Lynn, *et. al*
Phys. Rev. C **97** (2018)



I. Tews, R. Somasundaram, *et. al*
arXiv:2407.08979 (2024)

Quantum Monte Carlo

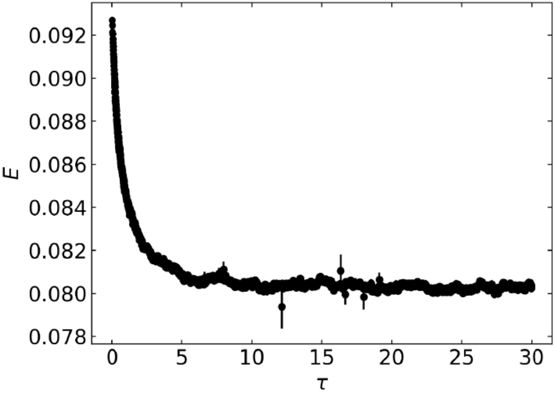
Schrodinger equation in imaginary time $$-\frac{\partial}{\partial \tau}\Psi = \left(-\frac{\hbar}{2m}\nabla^2 + V\right)\Psi$$

$$\psi(\tau) = e^{-(H-E_T)\tau}\psi_T$$

$$\psi(\tau) = \sum_{i=0}^{\infty} a_i e^{-(E_i - E_T)\tau}\psi_i$$

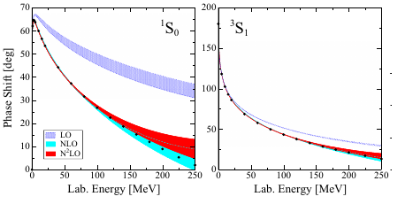
$$\psi(\tau) = a_0\psi_0 \qquad \lim \tau \to \infty$$

"Projects" out the ground-state energy and wavefunction



Local Chiral EFT

- QMC needs a local coordinate space interaction
- Use only half of the operators in calculation
- Recover the rest through anti-symmetrization



- $q \rightarrow r$ local
- $k \rightarrow \nabla$ non-local

$$V^{(0)} = \boxed{\alpha_1} + \boxed{\alpha_2 \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2} + \alpha_3 \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 + \alpha_4 \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2$$

$$\begin{aligned} V^{(2)}_{\text{cont}} = &\boxed{\gamma_1 q^2} + \boxed{\gamma_2 q^2 \boldsymbol{\sigma_1} \cdot \boldsymbol{\sigma_2}} + \boxed{\gamma_3 q^2 \boldsymbol{\tau_1} \cdot \boldsymbol{\tau_2}} \\ &+ \boxed{\gamma_4 q^2 \boldsymbol{\sigma_1} \cdot \boldsymbol{\sigma_2} \boldsymbol{\tau_1} \cdot \boldsymbol{\tau_2}} \\ &+ \gamma_5 k^2 + \gamma_6 k^2 \boldsymbol{\sigma_1} \cdot \boldsymbol{\sigma_2} + \gamma_7 k^2 \boldsymbol{\tau_1} \cdot \boldsymbol{\tau_2} \\ &+ \gamma_8 k^2 \boldsymbol{\sigma_1} \cdot \boldsymbol{\sigma_2} \boldsymbol{\tau_1} \cdot \boldsymbol{\tau_2} \\ &+ \boxed{\gamma_9 (\boldsymbol{\sigma_1} + \boldsymbol{\sigma_2})(\boldsymbol{q} \times \boldsymbol{k})} \\ &+ \gamma_{10} (\boldsymbol{\sigma_1} + \boldsymbol{\sigma_2})(\boldsymbol{q} \times \boldsymbol{k}) \boldsymbol{\tau_1} \cdot \boldsymbol{\tau_2} \\ &+ \boxed{\gamma_{11} (\boldsymbol{\sigma_1} \cdot \boldsymbol{q})(\boldsymbol{\sigma_2} \cdot \boldsymbol{q})} \\ &+ \boxed{\gamma_{12} (\boldsymbol{\sigma_1} \cdot \boldsymbol{q})(\boldsymbol{\sigma_2} \cdot \boldsymbol{q}) \boldsymbol{\tau_1} \cdot \boldsymbol{\tau_2}} \\ &+ \gamma_{13} (\boldsymbol{\sigma_1} \cdot \boldsymbol{k})(\boldsymbol{\sigma_2} \cdot \boldsymbol{k}) \\ &+ \gamma 14 (\boldsymbol{\sigma_1} \cdot \boldsymbol{k})(\boldsymbol{\sigma_2} \cdot \boldsymbol{k}) \boldsymbol{\tau_1} \cdot \boldsymbol{\tau_2} \end{aligned}$$

A. Gezerlis, I. Tews, E. Epelbaum, S. Gandolfi, K. Hebeler, A. Nogga, and A. Schwenk, Phys. Rev. Lett. **111**, 032501 (2013)

Trouble on the horizon ...

- 30 new contact operators at N^{3}LO
- Cannot avoid non-local operators

$$
\begin{aligned}
V_{\mathrm{cont}}^{(4)} = \; & \alpha_1\, \boldsymbol{q}^4 + \alpha_2\, \boldsymbol{q}^4 \boldsymbol{\tau}_1\cdot\boldsymbol{\tau}_2 + \alpha_3\, \boldsymbol{q}^4 \boldsymbol{\sigma}_1\cdot\boldsymbol{\sigma}_2 + \alpha_4\, \boldsymbol{q}^4 \boldsymbol{\sigma}_1\cdot\boldsymbol{\sigma}_2 \boldsymbol{\tau}_1\cdot\boldsymbol{\tau}_2 \\
& + \alpha_5\, \boldsymbol{k}^4 + \alpha_6\, \boldsymbol{k}^4 \boldsymbol{\tau}_1\cdot\boldsymbol{\tau}_2 + \alpha_7\, \boldsymbol{k}^4 \boldsymbol{\sigma}_1\cdot\boldsymbol{\sigma}_2 + \alpha_8\, \boldsymbol{k}^4 \boldsymbol{\sigma}_1\cdot\boldsymbol{\sigma}_2 \boldsymbol{\tau}_1\cdot\boldsymbol{\tau}_2 \\
& + \alpha_9\, \boldsymbol{q}^2\boldsymbol{k}^2 + \alpha_{10}\, \boldsymbol{q}^2\boldsymbol{k}^2 \boldsymbol{\tau}_1\cdot\boldsymbol{\tau}_2 + \alpha_{11}\, \boldsymbol{q}^2\boldsymbol{k}^2 \boldsymbol{\sigma}_1\cdot\boldsymbol{\sigma}_2 + \alpha_{12}\, \boldsymbol{q}^2\boldsymbol{k}^2 \boldsymbol{\sigma}_1\cdot\boldsymbol{\sigma}_2 \boldsymbol{\tau}_1\cdot\boldsymbol{\tau}_2 \\
& + \alpha_{13}\, (\boldsymbol{q}\times\boldsymbol{k})^2 + \alpha_{14}\, (\boldsymbol{q}\times\boldsymbol{k})^2 \boldsymbol{\tau}_1\cdot\boldsymbol{\tau}_2 + \alpha_{15}\, (\boldsymbol{q}\times\boldsymbol{k})^2 \boldsymbol{\sigma}_1\cdot\boldsymbol{\sigma}_2 + \alpha_{16}\, (\boldsymbol{q}\times\boldsymbol{k})^2 \boldsymbol{\sigma}_1\cdot\boldsymbol{\sigma}_2 \boldsymbol{\tau}_1\cdot\boldsymbol{\tau}_2 \\
& + \frac{i}{2}\alpha_{17}\, \boldsymbol{q}^2 (\boldsymbol{\sigma}_1+\boldsymbol{\sigma}_2)\cdot(\boldsymbol{q}\times\boldsymbol{k}) + \frac{i}{2}\alpha_{18}\, \boldsymbol{q}^2 (\boldsymbol{\sigma}_1+\boldsymbol{\sigma}_2)\cdot(\boldsymbol{q}\times\boldsymbol{k}) \boldsymbol{\tau}_1\cdot\boldsymbol{\tau}_2 \\
& + \frac{i}{2}\alpha_{19}\, \boldsymbol{k}^2 (\boldsymbol{\sigma}_1+\boldsymbol{\sigma}_2)\cdot(\boldsymbol{q}\times\boldsymbol{k}) + \frac{i}{2}\alpha_{20}\, \boldsymbol{k}^2 (\boldsymbol{\sigma}_1+\boldsymbol{\sigma}_2)\cdot(\boldsymbol{q}\times\boldsymbol{k}) \boldsymbol{\tau}_1\cdot\boldsymbol{\tau}_2 \\
& + \alpha_{21}\, \boldsymbol{q}^2\, \boldsymbol{\sigma}_1\cdot\boldsymbol{q}\, \boldsymbol{\sigma}_2\cdot\boldsymbol{q} + \alpha_{22}\, \boldsymbol{q}^2\, \boldsymbol{\sigma}_1\cdot\boldsymbol{q}\, \boldsymbol{\sigma}_2\cdot\boldsymbol{q}\, \boldsymbol{\tau}_1\cdot\boldsymbol{\tau}_2 + \alpha_{23}\, \boldsymbol{k}^2\, \boldsymbol{\sigma}_1\cdot\boldsymbol{q}\, \boldsymbol{\sigma}_2\cdot\boldsymbol{q} + \alpha_{24}\, \boldsymbol{k}^2\, \boldsymbol{\sigma}_1\cdot\boldsymbol{q}\, \boldsymbol{\sigma}_2\cdot\boldsymbol{q}\, \boldsymbol{\tau}_1\cdot\boldsymbol{\tau}_2 \\
& + \alpha_{25}\, \boldsymbol{q}^2\, \boldsymbol{\sigma}_1\cdot\boldsymbol{k}\, \boldsymbol{\sigma}_2\cdot\boldsymbol{k} + \alpha_{26}\, \boldsymbol{q}^2\, \boldsymbol{\sigma}_1\cdot\boldsymbol{k}\, \boldsymbol{\sigma}_2\cdot\boldsymbol{k}\, \boldsymbol{\tau}_1\cdot\boldsymbol{\tau}_2 + \alpha_{27}\, \boldsymbol{k}^2\, \boldsymbol{\sigma}_1\cdot\boldsymbol{k}\, \boldsymbol{\sigma}_2\cdot\boldsymbol{k} + \alpha_{28}\, \boldsymbol{k}^2\, \boldsymbol{\sigma}_1\cdot\boldsymbol{k}\, \boldsymbol{\sigma}_2\cdot\boldsymbol{k}\, \boldsymbol{\tau}_1\cdot\boldsymbol{\tau}_2 \\
& + \alpha_{29}\, \boldsymbol{\sigma}_1\cdot(\boldsymbol{q}\times\boldsymbol{k})\, \boldsymbol{\sigma}_2\cdot(\boldsymbol{q}\times\boldsymbol{k}) + \alpha_{30}\, \boldsymbol{\sigma}_1\cdot(\boldsymbol{q}\times\boldsymbol{k})\, \boldsymbol{\sigma}_2\cdot(\boldsymbol{q}\times\boldsymbol{k}) \boldsymbol{\tau}_1\cdot\boldsymbol{\tau}_2\,.
\end{aligned}
$$

- Need develop a way to self consistently treat non-local operators perturbatively

R. Somasundaram, J.E. Lynn, L. Huth, A. Schwenk, and I. Tews, Phys. Rev. C. **109**, 034005 (2024)

The problem?

First- and Second-order Corrections

$$E_0^{(1)} = \langle \psi_0 | V' | \psi_0 \rangle$$

$$E_0^{(2)} = -\sum_{k \neq 0}^{\infty} \frac{| \langle \psi_0 | V' \boxed{| \psi_k \rangle} |^2}{\boxed{E_k} - E_0}$$

The problem?

$$\lim_{\tau \to \infty} \psi(\tau) = \lim_{\tau \to \infty} e^{-(H - E_T)\tau} \psi_T \propto \psi_0 \longrightarrow E_0$$

A solution - In very brief

$$I(\mathcal{T}) = \int_0^{\mathcal{T}} d\tau \, \langle\psi_0| \, V' e^{-[H_0 - E_0]\tau} V' \, |\psi_0\rangle$$

$$I(\mathcal{T}) = (E_0^{(1)})^2 \mathcal{T} \boxed{- \sum_{k \neq 0}^{\infty} \frac{| \langle\psi_k|V'|\psi_0\rangle |^2}{E_k - E_0}} [\cancel{e^{-[E_k - E_0]\mathcal{T}}} - 1]$$

$$I(\mathcal{T} \to \infty) = (E_0^{(1)})^2 \mathcal{T} - E_0^{(2)}$$

R. Curry, J.E. Lynn, K.E. Schmidt and A. Gezerlis, Phys. Rev. Res., **5**, L042021 (2023)

Non-Local Operators

- Non-local operators cannot be avoided at higher chiral orders.
- Non-local operators cannot easily be included in imaginary time propagator
- Let's replace a local operator at NLO with a non-local equivalent and treat perturbatively

- $q \to r$ local
- $k \to \nabla$ non-local

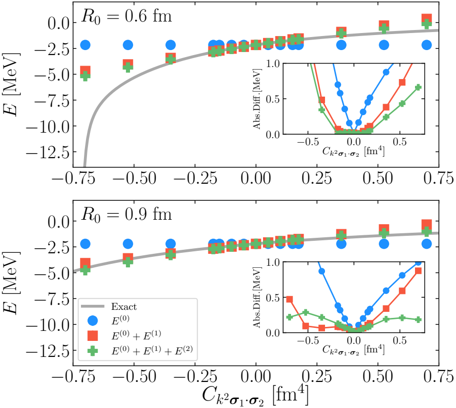
$$
\begin{aligned}
V^{(0)} &= \alpha_1 + \alpha_2 \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 + \alpha_3 \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \\
&\quad + \alpha_4 \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2
\end{aligned}
$$

$$
\begin{aligned}
V^{(2)}_{\text{cont}} = & \boxed{\gamma_1 q^2} + \boxed{\gamma_2 q^2 \boldsymbol{\sigma_1} \cdot \boldsymbol{\sigma_2}} + \boxed{\gamma_3 q^2 \boldsymbol{\tau_1} \cdot \boldsymbol{\tau_2}} \\
& + \boxed{\gamma_4 q^2 \boldsymbol{\sigma_1} \cdot \boldsymbol{\sigma_2} \boldsymbol{\tau_1} \cdot \boldsymbol{\tau_2}} \\
& + \gamma_5 k^2 + \boxed{\gamma_6 k^2 \boldsymbol{\sigma_1} \cdot \boldsymbol{\sigma_2}} + \gamma_7 k^2 \boldsymbol{\tau_1} \cdot \boldsymbol{\tau_2} \\
& + \gamma_8 k^2 \boldsymbol{\sigma_1} \cdot \boldsymbol{\sigma_2} \boldsymbol{\tau_1} \cdot \boldsymbol{\tau_2} \\
& + \boxed{\gamma_9 (\boldsymbol{\sigma_1} + \boldsymbol{\sigma_2})(\boldsymbol{q} \times \boldsymbol{k})} \\
& + \gamma_{10} (\boldsymbol{\sigma_1} + \boldsymbol{\sigma_2})(\boldsymbol{q} \times \boldsymbol{k}) \boldsymbol{\tau_1} \cdot \boldsymbol{\tau_2} \\
& + \boxed{\gamma_{11} (\boldsymbol{\sigma_1} \cdot \boldsymbol{q})(\boldsymbol{\sigma_2} \cdot \boldsymbol{q})} \\
& + \boxed{\gamma_{12} (\boldsymbol{\sigma_1} \cdot \boldsymbol{q})(\boldsymbol{\sigma_2} \cdot \boldsymbol{q}) \boldsymbol{\tau_1} \cdot \boldsymbol{\tau_2}} \\
& + \gamma_{13} (\boldsymbol{\sigma_1} \cdot \boldsymbol{k})(\boldsymbol{\sigma_2} \cdot \boldsymbol{k}) \\
& + \gamma 14 (\boldsymbol{\sigma_1} \cdot \boldsymbol{k})(\boldsymbol{\sigma_2} \cdot \boldsymbol{k}) \boldsymbol{\tau_1} \cdot \boldsymbol{\tau_2}
\end{aligned}
$$

R. Curry, R. Somasundaram, S. Gandolfi, A. Gezerlis, and I. Tews, Phys. Rev. C, **111**, 015801 (2025)

Deuteron with non-local operator

$$k^2 \longrightarrow \left[-\frac{1}{4}\left(\nabla^2\delta_{R_0}\psi(\boldsymbol{r}) - \frac{1}{r}\frac{\partial\delta_{R_0}}{\partial r}\left(\boldsymbol{r}\cdot\nabla\psi(\boldsymbol{r})\right) - \delta_{R_0}\nabla^2\psi(\boldsymbol{r}\right)\right]$$

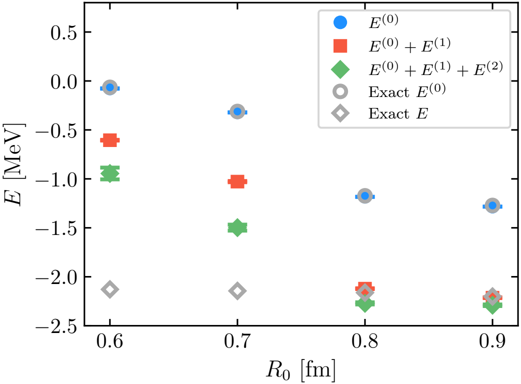


- Non-local operator not included in fit to phase shifts
- $k^2\boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2$ included perturbatively for two cutoffs
- Compare against exact Lippmann-Schwinger solution

R. Curry, R. Somasundaram, S. Gandolfi, A. Gezerlis, and I. Tews, Phys. Rev. C, **111**, 015801 (2025)

A more realistic test case

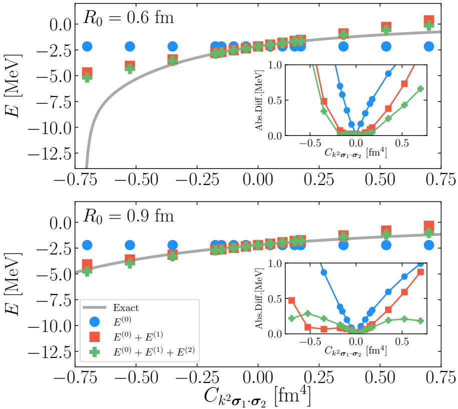
- Non-local operator <u>included</u> in fit to phase shifts

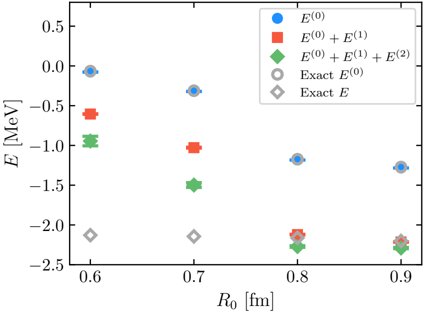


- Full range of coordinate space cutoffs
- Issues with perturbativeness at harder cutoffs

R. Curry, R. Somasundaram, S. Gandolfi, A. Gezerlis, and I. Tews, Phys. Rev. C, **111**, 015801 (2025)

Wait a minute...

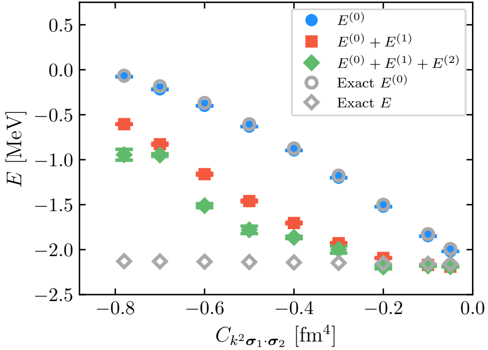




Can we take inspiration from the first case, and improve on the hard cutoff?

R. Curry, R. Somasundaram, S. Gandolfi, A. Gezerlis, and I. Tews, Phys. Rev. C, **111**, 015801 (2025)

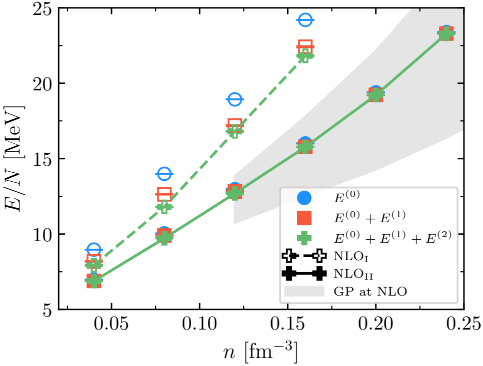
Constraining the non-local operator



- Non-local operator <u>included</u> in fit to phase shifts
- Create a series of NLO interactions with a variable non-local LEC
- Clear region where perturbation theory is sufficient!

<u>R. Curry</u>, R. Somasundaram, S. Gandolfi, A. Gezerlis, and I. Tews, Phys. Rev. C, **111**, 015801 (2025)

Neutron Matter



- Use constrained NLO interaction for calculation of neutron matter EOS (see poster for details)
- Clear improvement from constraining the non-local operator
- Able to include non-local operators perturbatively even for hard cutoff interactions!

R. Curry, R. Somasundaram, S. Gandolfi, A. Gezerlis, and I. Tews, Phys. Rev. C, **111**, 015801 (2025)

I. Tews, R. Somasundaram, *et. al* arXiv:2407.08979

Acknowledgements

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- **Alex Gezerlis** (Guelph)
- **Ingo Tews** (LANL)
- **Stefano Gandolfi** (LANL)
- **Rahul Somasundaram** (LANL, SU)
- **Kevin Schmidt** (ASU)
- **Joel Lynn** (Darmstadt)

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Los Alamos NATIONAL LABORATORY

Digital Research Alliance of Canada

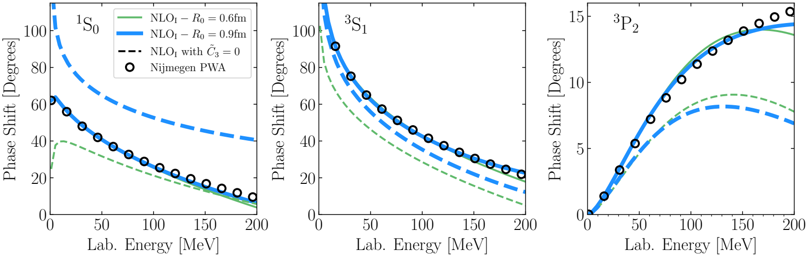
INNOVATION Canada Foundation for Innovation Fondation canadienne pour l'innovation

Summary

- Extend QMC perturbation theory formalism to handle non-local operators for realistic nuclear interactions
- Successfully constrained non-local operator to treat perturbatively for hard-cutoff interactions
- Paves the way for continuum AFDMC calculations of light nuclei and the neutron matter EOS at N^3LO

Thank you for your attention!

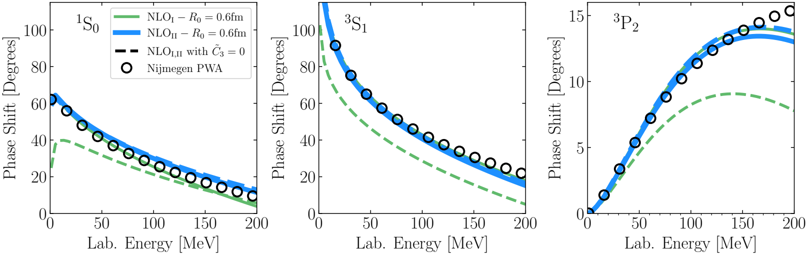
Phase Shifts with and without the nonlocal operator



- Missing physics at the two-body level, but add it perturbatively at the many-body level

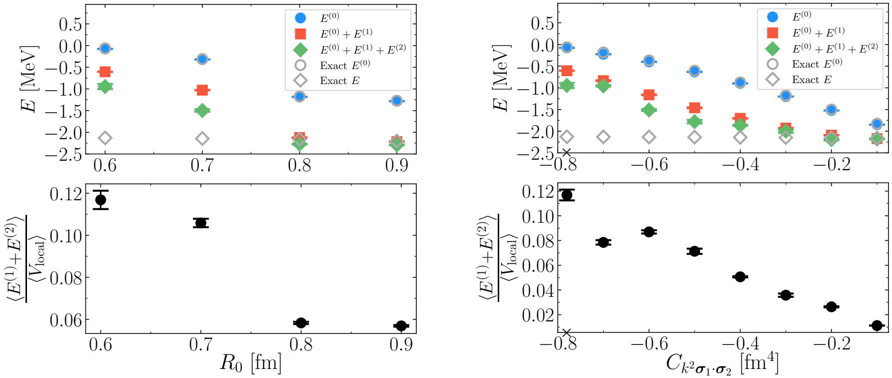
Phase Shifts with and without the nonlocal operator

- We can look at the phase shifts for our "constrained" interaction



- We can see at the two-body level that we are missing much less of the physics

Measure of Perturbativeness



Spin / Isospin Complications

DMC is very good for central interactions

$$e^{-\sum_{i<j} v(\boldsymbol{r}_{ij})\tau}$$

How do we handle a propagator with spin/isospin degrees of freedom?

$$e^{-\sum_{i<j}[v_1(\boldsymbol{r}_{ij})\boldsymbol{\sigma}_1\cdot\boldsymbol{\sigma}_2+v_2(\boldsymbol{r}_{ij})\boldsymbol{\tau}_1\cdot\boldsymbol{\tau}_2+\ldots]\tau}$$

The spin-spin term is equivalent to a Majorana exchange operator,

$$\boldsymbol{\sigma}_1\cdot\boldsymbol{\sigma}_2 = 2P_{12}^{\boldsymbol{\sigma}} - 1$$

Green's Function Monte Carlo

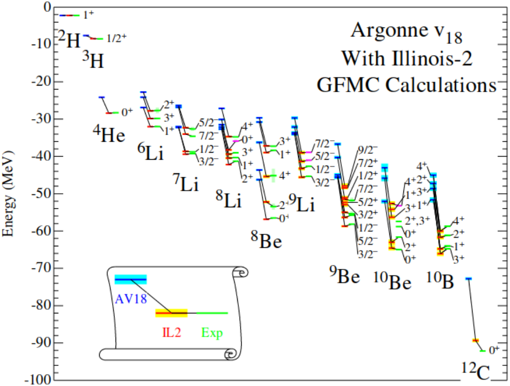
- Explicitly sum over all spin/isospin states

$$|\Phi\rangle = \begin{pmatrix} a_{\uparrow\uparrow\uparrow} \\ a_{\uparrow\uparrow\downarrow} \\ a_{\uparrow\downarrow\uparrow} \\ a_{\uparrow\downarrow\downarrow} \\ a_{\downarrow\uparrow\uparrow} \\ a_{\downarrow\uparrow\downarrow} \\ a_{\downarrow\downarrow\uparrow} \\ a_{\downarrow\downarrow\downarrow} \end{pmatrix} \qquad \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 |\Phi\rangle = \begin{pmatrix} a_{\uparrow\uparrow\uparrow} \\ a_{\uparrow\uparrow\downarrow} \\ 2a_{\downarrow\uparrow\uparrow} - a_{\uparrow\downarrow\uparrow} \\ 2a_{\downarrow\uparrow\downarrow} - a_{\uparrow\downarrow\downarrow} \\ 2a_{\uparrow\downarrow\uparrow} - a_{\downarrow\uparrow\uparrow} \\ 2a_{\uparrow\downarrow\downarrow} - a_{\downarrow\uparrow\downarrow} \\ a_{\downarrow\downarrow\uparrow} \\ a_{\downarrow\downarrow\downarrow} \end{pmatrix}$$

- Each of the $2^A \frac{A!}{Z!(A-Z)!}$ spin-isospin states must be propagated
- Limiting GFMC to $A \leq 12$

J. Carlson, Phys. Rev. C **36** (1987)

Green's Function Monte Carlo



S. Pieper, R.B. Wiringa,
Annu. Rev. Nucl. Part. Sci. **51** (2001)

- GFMC is an exact calculation
- Limited to $A \leq 12$
- Still used to study electroweak observables,
  G.B. King and S. Pastore, Annu. Rev. Nucl. Part. Sci. **74** (2024)
- and few-body nuclear reactions
  J.E. Lynn, I. Tews, J. Carlson, *et. al*, Phys. Rev. Lett., **116** (2016)

Auxiliary Field Diffusion Monte Carlo

- Instead of summing over all spin-isospin states, we can sample them
- Particles can be described by single-nucleon spinors

$$|S_i\rangle = a_i\,|p\uparrow\rangle + b_i\,|p\downarrow\rangle + c_i\,|n\uparrow\rangle + d_i\,|n\downarrow\rangle$$

- Linearize quadratic spin/isospin ($\boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2, etc$) dependence through Hubbard-Stratonovich Transformation

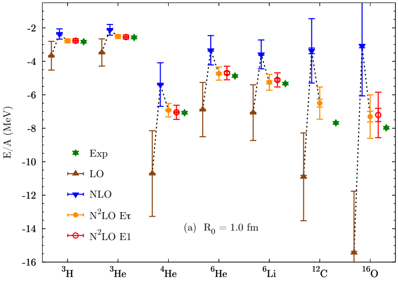
$$e^{-\frac{1}{2}\lambda O^2} = \frac{1}{\sqrt{2\pi}} \int dx_e^{-\frac{x^2}{2}+\sqrt{-\lambda}xO}$$

- We have traded two body $\boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2$ type interactions for single particles interacting with external auxiliary fields x

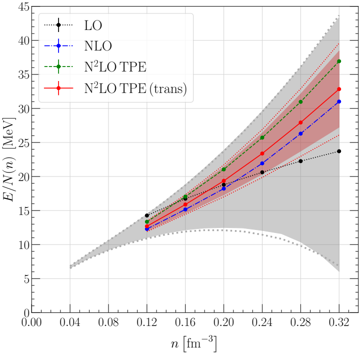
K.E. Schmidt and S. Fantoni, Phys. Lett. B **446** (1999)

Auxiliary Field Diffusion Monte Carlo

- Trade off in precision due to sampling auxiliary fields
- Extend calculations to $A \leq 18$ as well as dense matter



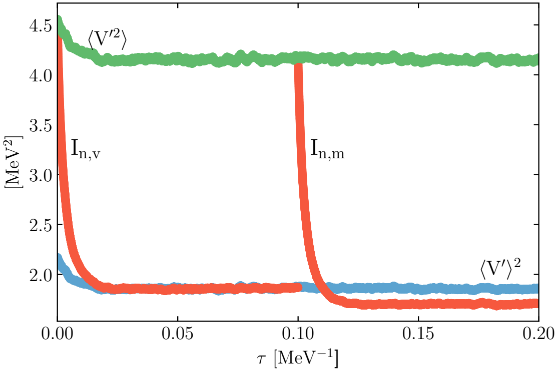
D. Lonardoni, S. Gandolfi, J.E. Lynn, *et. al*
Phys. Rev. C **97** (2018)



I. Tews, R. Somasundaram, *et. al*
arXiv:2407.08979 (2024)

Practically speaking

$$I(\mathcal{T}) = \int_0^{\mathcal{T}} \boxed{d\tau \langle\psi_0| V' e^{-[H_0-E_0]\tau} V' |\psi_0\rangle} \longrightarrow I \approx \frac{\sum_i^{\mathcal{N}} w_i V'(\boldsymbol{R}_i) V'(\boldsymbol{R}_i')}{\sum_i^{\mathcal{N}} w_i}$$



R. Curry, J.E. Lynn, K.E. Schmidt and A. Gezerlis, Phys. Rev. Res., **5**, (2023)